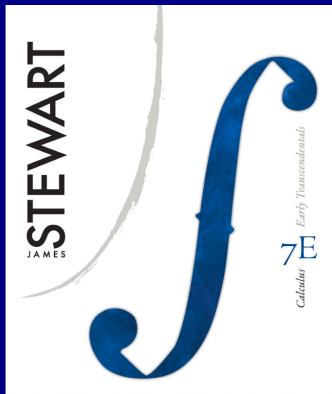




MATH1560J RC4

1. Integral

Definition

A function F is called an antiderivative of f on an interval I if $F'(x) = f(x)$ for all x in I .

Theorem

If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is

$$F(x) + C$$

, where C is an arbitrary constant.

Function	Antiderivative	Function	Antiderivative
$cf(x)$	$cF(x)$	$\sec^2 x$	$\tan x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sec x \tan x$	$\sec x$
$x^n (n \neq -1)$	$\frac{x^{n+1}}{n+1}$	$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
$\frac{1}{x}$	$\ln x $	$\frac{1}{1+x^2}$	$\tan^{-1} x$
e^x	e^x	$\cosh x$	$\sinh x$
$\cos x$	$\sin x$	$\sinh x$	$\cosh x$
$\sin x$	$-\cos x$		

Notation

$$\int f(x)dx = F(x) \quad \text{means} \quad F'(x) = f(x)$$

How to understand this notation?

Theorem

If the antiderivatives of functions $f(x)$ and $g(x)$ exist, then for any constant p and q , the antiderivate of $pf(x)+qg(x)$ also exists, and we have

$$\int [pf(x) + qg(x)]dx = p \int f(x)dx + q \int g(x)dx$$

Keep the following indefinite integrals firmly in mind!

$$\int cf(x)dx = c \int f(x)dx$$

$$\int kdx = kx + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \frac{1}{x^2 + 1} dx = \arctan x + C$$

$$\int \sinh x dx = \cosh x + C$$

$$\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

$$\int \cosh x dx = \sinh x + C$$

Definition

$$\int f(x)g'(x)dx = \int f(x)dg(x)$$

How to memorize?

$$\frac{dg(x)}{dx} = g'(x)$$

$$\Rightarrow dx = \frac{dg(x)}{g'(x)}$$

$$\Rightarrow \int f(x)g'(x)dx = \int f(x)g'(x) \cdot \frac{dg(x)}{g'(x)} = \int f(x)dg(x)$$

Type 1: Direct Substitution ($u = g(x)$)

Substitution Rule for Direct Substitution

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I , then

$$\int f(g(x))g'(x)dx = \int f(g(x))dg(x) = \int f(u)du = F(u) + C$$

Type 2: Inverse Substitution ($x = \varphi(t)$)

Substitution Rule for Inverse Substitution

If $x = \varphi(t)$ is an invertible function, then

$$\begin{aligned}\int f(x)dx &= \int f(\varphi(t))d\varphi(t) = \int f(\varphi(t))\varphi'(t)dt \\ &= \tilde{F}(t) = \tilde{F}(\varphi^{-1}(x)) + C\end{aligned}$$

$$\textcircled{1} \int \frac{3x^3}{1-x^4} dx$$

$$\textcircled{2} \int \frac{\ln x}{x} dx$$

$$\textcircled{3} \int \frac{\sec^2 x}{\tan^{10} x} dx$$

$$\textcircled{1} \text{ ans} = 0.75 \int \frac{1}{1-x^4} dx^4 = -\frac{3}{3} \int \frac{1}{1-x^4} d(1-x^4) = -\frac{3}{4} \ln|1-x^4| + C$$

$$\textcircled{2} d(\ln x) = dx \cdot \frac{1}{x} \Rightarrow \text{ans} = \int \ln x d(\ln x) = \frac{1}{2} \ln^2 x + C$$

$$\textcircled{3} d(\tan x) = \sec^2 x \cdot dx \Rightarrow \text{ans} = \int \frac{d(\tan x)}{\tan^{10} x} = -\frac{1}{9} \tan^{-9} x + C$$

An Important Application: Trigonometric Substitutions:

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, \quad 0 \leq \theta < \frac{\pi}{2}$	$\sec^2 \theta - 1 = \tan^2 \theta$

$$\int \frac{1}{x^2 \sqrt{x^2 + 4}} dx$$

Exercise 2:

Let $x =: 2\tan\theta$, $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $\frac{dx}{d\theta} = 2\sec^2\theta$, $\sqrt{x^2 + 4} = 2\sec\theta$, $\text{ans} = \int \frac{\cos\theta}{4\sin^2\theta} d\theta$.

Then $u := \sin\theta$, $\text{ans} = \int \frac{du}{4u^2} = -\frac{1}{4u} + C = -\frac{1}{4\sin\theta} + C = \frac{-\sqrt{x^2 + 4}}{4x} + C$

Suppose f is continuous on $[-a, a]$, $a > 0$.

If f is even, i.e., $\forall x \ f(x) = f(-x)$, then

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

If f is odd, i.e., $\forall x \ f(x) = -f(-x)$, then

$$\int_{-a}^a f(x) dx = 0$$

Assume a function f , continuous on $[a,b]$, then

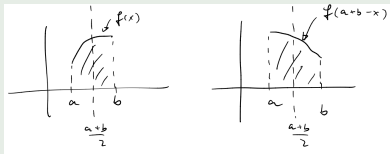
$$I = \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

Let $y = a + b - x$, so $\frac{dy}{dx} = -1$

$$\int_a^b f(y) dy = \int_b^a \underbrace{f(a+b-x)}_{y(x)} \frac{dy}{dx} dx$$

$$\Rightarrow 2I = \int_a^b f(x) dx + \int_a^b f(a+b-x) dx$$

$$\Rightarrow I = \frac{1}{2} \int_a^b [f(x) + f(a+b-x)] dx$$



Let function $f : \mathbb{R} \rightarrow \mathbb{R}$, continuous, periodic w/ period $T > 0$, i.e.,
 $f(x) = f(x + T) \quad \forall x \in \mathbb{R} \text{ then } \forall n \in \mathbb{N} \quad (n = 0, 1, 2, \dots)$

$$\int_0^{nT} f(x) dx = n \int_0^T f(x) dx$$

$$\int u dv = uv - \int v du$$

$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

How to choose $f(x)$ and $g'(x)$?

Among the following function types:

the more to the left, the more suitable to be $f(x)$

the more to the right, the more suitable to be $g'(x)$

(left) inverse trigonometric function, logarithm function, power function, trigonometric function, exponential function **(right)**

When should we consider applying Integration by Parts?

- ① If the integrand is the product of **inverse trigonometric function, logarithm function or power function** and another function **whose antiderivative is easy to calculate**.
- ② If the integrand is the product of **trigonometric function and exponential function**, apply Integration by Part twice in order to find an identical equation about the original integral, and then solve that equation.
- ③ If the integrand contains n (or a relatively large number), apply Integration by Part to **find the recursion formula**.

① $\int x \sin x \, dx$

② $\int \ln^2 x \, dx$

③ $\int x^2 e^x \, dx$

Exercise 3:

$$\textcircled{1} \text{ ans} = - \int x d(\cos x) = -x \cos x + \int \cos x dx = -x \cos x + \sin x + C$$

$$\textcircled{2} \text{ ans} = x \ln^2 x - \int 2 \ln x dx = x \ln^2 x - 2x \ln x + \int 2 dx = x \ln^2 x - 2x \ln x + 2x + C$$

$$\textcircled{3} \text{ ans} = x^2 e^x - 2 \int x e^x dx = x^2 e^x - 2x e^x + 2 \int e^x dx = (x^2 - 2x + 2) e^x + C$$

- ① If at least m and n is odd, adopt substitution rule.

$$\int \sin^m x \cos^{2n+1} x \, dx = \int \sin^m x \cos^{2n} x \, d(\sin x) =$$

$$\int \sin^m x \cdot (1 - \sin^2 x)^n \, d(\sin x) \stackrel{u=\sin x}{=} \int u^m (1 - u^2)^n \, du$$

$$\begin{aligned} \int \cos^m x \sin^{2n+1} x \, dx &= - \int \cos^m x \sin^{2n} x \, d(\cos x) = \\ &= - \int \cos^m x \cdot (1 - \cos^2 x)^n \, d(\cos x) \stackrel{u=\cos x}{=} - \int u^m (1 - u^2)^n \, du \end{aligned}$$

- ② Else, use half-angle identities that $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$,
 $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$.

- ① If m is even (not 0), use $d(\tan x) = \sec^2 x dx$ to transform it into the integration of $\tan x$.

$$\int \sec^{2k} x \tan^n x dx = \int (\tan^2 x + 1)^k \tan^n x d(\tan x) \stackrel{u=\tan x}{=} \int (u^2 + 1)^k u^n du.$$

- ② If n is odd, use $d(\sec x) = \sec x \tan x dx$ to transform it into the integration of $\sec x$.

$$\int \tan^{2k+1} x \sec^n x dx = \int (\sec^2 x - 1)^k \sec^{n-1} x d(\sec x) = \dots$$

Though not as useful, you can also explore $\int \csc^m x \cot^n x dx$.

Use Product-to-Sum formula:

$$\textcircled{1} \sin \alpha \cos \beta = \frac{1}{2}(\sin(\alpha + \beta) + \sin(\alpha - \beta))$$

$$\textcircled{2} \cos \alpha \sin \beta = \frac{1}{2}(\sin(\alpha + \beta) - \sin(\alpha - \beta))$$

$$\textcircled{3} \sin \alpha \sin \beta = \frac{1}{2}(\cos(\alpha + \beta) + \cos(\alpha - \beta))$$

$$\textcircled{4} \cos \alpha \cos \beta = -\frac{1}{2}(\cos(\alpha + \beta) - \cos(\alpha - \beta))$$

① $\int_0^{\pi/2} \sin^7 x \cos^5 x \, dx$

Exercise 4:

$$\textcircled{1} \int_0^{\pi/2} \sin^7 x \cos^5 x \, dx = \int_0^1 \sin^7 x (1 - \sin^2 x)^2 d(\sin x) = \left[\frac{1}{8} u^8 - \frac{1}{5} u^{10} + \frac{1}{12} u^{12} \right]_0^1 = \frac{1}{120}$$

Use **induction**.

$$\begin{aligned}
 & \int e^{ax} \sin^n x \, dx \\
 &= \frac{1}{a} \int \sin^n x \, d(e^{ax}) \\
 &= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a} \int e^{ax} \sin^{n-1} x \cos x \, dx \\
 &= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} e^{ax} \sin^{n-1} x \cos x + \frac{n}{a^2} \int e^{ax} ((n-1) \sin^{n-2} x \cos^2 x - \sin^n x) \, dx \\
 &= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} e^{ax} \sin^{n-1} x \cos x + \frac{n}{a^2} \int e^{ax} ((n-1) \sin^{n-2} x - n \sin^n x) \, dx
 \end{aligned}$$

$\Rightarrow$

$$\frac{a^2 + n^2}{a^2} \int e^{ax} \sin^n x \, dx = \frac{e^{ax} \sin^{n-1} x}{a} \left(\sin x - \frac{n}{a} \cos x \right) + \frac{n(n-1)}{a^2} \int e^{ax} \sin^{n-2} x \, dx$$

$\Rightarrow$

$$\int e^{ax} \sin^n x \, dx = \frac{e^{ax} \sin^{n-1} x}{a^2 + n^2} (a \sin x - n \cos x) + \frac{n(n-1)}{a^2 + n^2} \int e^{ax} \sin^{n-2} x \, dx$$

Use similar method,

$$\begin{aligned} \textcircled{1} \quad & \int e^{ax} \sin^n x \, dx \\ &= \frac{e^{ax} \cos^{n-1} x}{a^2 + n^2} (a \cos x + n \sin x) + \frac{n(n-1)}{a^2 + n^2} \int e^{ax} \cos^{n-2} x \, dx \end{aligned}$$

$$\textcircled{2} \quad \int \sin^n x \, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

$$\textcircled{3} \quad \int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

$$\textcircled{4} \quad \int \sin^{-n} x \, dx = -\frac{1}{n-1} \cdot \frac{\cos x}{\sin^{n-1} x} + \frac{n-2}{n-1} \cdot \int \sin^{-(n-2)} x \, dx$$

$$\textcircled{5} \quad \int \cos^{-n} x \, dx = \frac{1}{n-1} \cdot \frac{\sin x}{\cos^{n-1} x} + \frac{n-2}{n-1} \cdot \int \cos^{-(n-2)} x \, dx$$

- ① Try to prove that

$$\int (\sec x)^{2n+1} dx = \frac{\tan x \cdot (\sec x)^{2n-1}}{2n} + \frac{2n-1}{2n} \int (\sec x)^{2n-1} dx \text{ if } n \geq 1$$

- ② Calculate $\int \sec^3 x dx$

Solution:

(1)

$$\begin{aligned} & \int (\sec x)^{2n+1} dx \\ &= \int (\sec x)^{2n-1} d(\tan x) \\ &= \tan x \cdot (\sec x)^{2n-1} - (2n-1) \int \tan^2 x \sec^{2n-1} x dx \\ &= \tan x \cdot (\sec x)^{2n-1} - (2n-1) \int \sec^{2n-1} x dx + (2n-1) \int \sec^{2n+1} x dx \end{aligned}$$

(2)

$$\frac{1}{2} (\ln |\tan x + \sec x| + \sec x \tan x) + C$$

Theorem

The antiderivatives of rational function are always elementary functions.

Here, rational functions are in the form that $\frac{p_m(x)}{q_n(x)}$, where $p_m(x)$ and $q_n(x)$ are polynomials in the degree of m and n .

Basic rational functions

$$① \int \frac{dx}{ax+b} = \frac{1}{a} \ln|ax+b| + C$$

$$② \int \frac{dx}{(ax+b)^n} = \int \frac{d(ax+b)}{a(ax+b)^n} = \frac{1}{a(n-1)(ax+b)^{n-1}} + C$$

$$③ \int \frac{xdx}{ax^2+bx+c} = \frac{1}{2a} \ln|ax^2+bx+c| - \frac{b}{2a} \int \frac{dx}{ax^2+bx+c}$$

$$④ \int \frac{dx}{ax^2+bx+c} \stackrel{\Delta=b^2-4ac<0}{=} \frac{2}{\sqrt{-\Delta}} \arctan \frac{2ax+b}{\sqrt{4ac-b^2}} + C$$

$$⑤ \int \frac{dx}{ax^2+bx+c} \stackrel{\Delta=b^2-4ac>0}{=} \frac{1}{\sqrt{\Delta}} \ln \left| \frac{2ax+b-\sqrt{\Delta}}{2ax+b+\sqrt{\Delta}} \right| + C$$

Remark. What if $\Delta = 0$?

Then, what about $\int \frac{x+d}{(ax^2+bx+c)^n} dx$?

$$\textcircled{1} \int \frac{x+d}{(ax^2+bx+c)^n} dx = \frac{1}{2a} \int \frac{2ax+b}{(ax^2+bx+c)^n} dx + (d - \frac{b}{2a}) \int \frac{1}{(ax^2+bx+c)^n} dx$$

$$\textcircled{2} \frac{1}{2a} \int \frac{2ax+b}{(ax^2+bx+c)^n} dx = \frac{1}{2a} \int \frac{d(ax^2+bx+c)}{(ax^2+bx+c)^n} = \dots$$

$$\textcircled{3} \int \frac{1}{(ax^2+bx+c)^n} dx = \frac{1}{a} \int \frac{dx}{((2ax+b)^2 - b^2 + 4ac)^n} = \frac{1}{8a} \int \frac{du}{(u^2 - b^2 + 4ac)^n}$$

$$\textcircled{4} \int \frac{du}{(au^2+b)^n} = \frac{2n-3}{2b(n-1)} \int \frac{du}{(au^2+b)^{n-1}} + \frac{u}{2b(n-1)(au^2+b)^{n-1}}$$

$\textcircled{5}$ Combine all tools above up!

Form:
$$\frac{f(x)}{[A(x)]^a[B(x)]^b\ldots} = \frac{f_a(x)}{[A(x)]^a} + \frac{f_b(x)}{[B(x)]^b} + \ldots$$

Purpose: Split a complex rational function into several functions whose forms are familiar and easy to integrate, or change the integration into familiar forms with trigonometric substitution

Critical skill: **Undetermined coefficient method, Trigonometrical Substitution, Reciprocal Substitution.**

Also, as rational functions are easier to deal with, we can carry out **Rationalization.**

Key character: Whether the denominator of the integrand can be factorized or not

If the degree of the polynomial (or other type of functions) in the numerator is greater than or equal to that in the denominator, **extract a polynomial (or other type of functions)**.

If the degree of the polynomial in the denominator is too high, in order to reduce the workload, it's recommended to **apply substitution rule before attempting to split the function**.

Exercise 6-Undetermined coefficient method

$$\textcircled{1} \int \frac{4y^2 - 7y - 12}{y(y+2)(y+3)} dy$$

$$\textcircled{2} \int \frac{x^4 + x^3 + 3x^2 - 1}{(x^2 + 1)^2(x - 1)} dx$$

Exercise 7-Reciprocal Substitution

$$\textcircled{1} \int \frac{dx}{x^8(1+x^2)}$$

Exercise 8-Rationalization

$$\textcircled{1} \int \frac{dx}{\sqrt{1-x} + \sqrt[3]{1-x}}$$

When encountering complex combination of trigonometric functions, we can substitute $t = \tan \frac{x}{2}$.

Then there is

$$\sin x = \frac{2t}{t^2 + 1}, \cos x = \frac{1 - t^2}{t^2 + 1}, \tan x = \frac{2t}{1 - t^2}, \frac{dx}{dt} = \frac{2}{t^2 + 1}.$$

It is thus converted into rational functions.

Exercise 9-"Universal Substitution"

① $\int \frac{dx}{1 + \cos x + \sin^2(\frac{x}{2})}$

Definition

If a function $f(x)$ is defined on the interval $[a, b]$, insert several points randomly, and we will get:

$$a = x_0 < x_1 < \cdots < x_n = b$$

which defines n intervals whose corresponding lengths are

$$\Delta x_k = x_k - x_{k-1}$$

The for every interval select $\xi_k \in [x_{k-1}, x_k]$, the Riemann Sum of $f(x)$ on the interval $[a, b]$ is:

$$S_n = \sum_{k=1}^n f(\xi_k) \Delta_k$$

Riemann Sum is the sum of areas of rectangles.

In order to calculate the area of a figure:

- ① Cut that irregular figure into small strips, and then regard the strip as a rectangle approximately.
- ② Measure the lengths of these small rectangles respectively and calculate each of their area.
- ③ Add up all the rectangular areas to get the total area.

Only applicable to one situation: $\int e^{x^2} dx$.

For simplicity, let us suggest $x \geq 0$.

$$\int e^{x^2} dx = \sqrt{\int e^{x^2} dx \cdot \int e^{y^2} dy} = \sqrt{\iint e^{x^2+y^2} dx dy} = \sqrt{\iint r e^{r^2} dr d\theta} = \sqrt{\pi e^{x^2}}.$$

Example 1

$$\int \sin(\ln x) dx =$$

$$x \sin(\ln x) - \int x \cos(\ln x) \cdot \frac{1}{x} dx = x \sin(\ln x) - x \cos(\ln x) - \int \sin(\ln x) dx$$

$$\Rightarrow \int \sin(\ln x) dx = \frac{x \sin(\ln x) - x \cos(\ln x)}{2}$$

Example 2

$$I = \int \frac{\sin^2 x}{\sin x + \sqrt{3} \cos x} dx$$

$$\text{define } J := \frac{\cos^2 x}{\sin x + \sqrt{3} \cos x} dx, I + J = \int \frac{1}{\sin x + \sqrt{3} \cos x} dx =$$

$$\frac{1}{2} \int \frac{1}{\sin(x + \frac{\pi}{3})} dx = \frac{1}{2} \ln \left| \csc(x + \frac{\pi}{3}) - \cot(x + \frac{\pi}{3}) \right| + C$$

$$I - 3J = \int \sin x - \sqrt{3} \cos x dx = -\cos x - \sqrt{3} \sin x + C$$

$$\text{Thus } I = \frac{3}{8} \ln \left| \csc(x + \frac{\pi}{3}) - \cot(x + \frac{\pi}{3}) \right| - \frac{1}{4} \cos x - \frac{\sqrt{3}}{4} \sin x + C$$

Definition

If f is a function defined for $a \leq x \leq b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$. We let $x_0(= a), x_1, x_2, \dots, x_n(= b)$ be the endpoints of these subintervals and we let $x_1^*, x_2^*, \dots, x_n^*$ be any sample points in these subintervals, so x_i^* lies in the i th subinterval. Then the definite integral of f from a to b is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

provided that this limit exists and gives the same value for all possible choices of sample points. If it does exist, we say that f is integrable on $[a, b]$.

Definite integral can also be intuitively understood as the net area between the curve $(f(x), x)$, line $x = a$, $x = b$, and x axis.

Properties

1 Linearity

$$\int_a^b [k_1 f(x) + k_2 g(x)] dx = k_1 \int_a^b f(x) dx + k_2 \int_a^b g(x) dx$$

2 Internal Additivity

$$\int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$$

3 Comparison Properties

If $f(x) \geq g(x)$ for $a \leq x \leq b$, then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$

If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

Properties

4 Absolute Integrability

If $f(x)$ is intergrable on $[a,b]$, then $|f(x)|$ is also integrable on $[a,b]$, and we have

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

Note

If $f(x)$ and $g(x)$ are intergrable on $[a,b]$, then $f(x) \cdot g(x)$ is also integrable on $[a,b]$, but generally

$$\int_a^b f(x)g(x)dx \neq \left(\int_a^b f(x)dx \right) \cdot \left(\int_a^b g(x)dx \right)$$

Calculate the definite integral by definition

① $\int_{-3}^0 (1 + \sqrt{9 - x^2}) dx$

② $\int_{\pi}^{\pi} \sin^2 x \cos^4 x dx$

Solutions:

- ① $\int_{-3}^0 (1 + \sqrt{9 - x^2}) dx$ can be interpreted as the area under the graph of $f(x) = 1 + \sqrt{9 - x^2}$ between $x = -3$ and $x = 0$. This is equal to one-quarter the area of the circle with radius 3, plus the area of the rectangle, so

$$\int_{-3}^0 (1 + \sqrt{9 - x^2}) dx = \frac{1}{4}\pi \cdot 3^2 + 1 \cdot 3 = 3 + \frac{9}{4}\pi$$

- ② $\int_{\pi}^{\pi} \sin^2 x \cos^4 x dx = 0$
since the limits of integral are equal

Newton-Leibniz Formula

Suppose f is continuous on $[a,b]$.

- If $g(x) = \int_a^x f(t)dt$, then $g'(x) = f(x)$
- If F is any antiderivative of f (i.e. $F' = f$), then
$$\int_a^b f(x)dx = F(b) - F(a)$$

$$1 \quad \int_0^{\pi} \sqrt{\sin^3 x - \sin^5 x} dx$$

$$2 \quad \int_0^3 \frac{x^2}{(x^2 - 3x + 3)^2} dx$$

Solutions: 1. $\int_0^\pi \sqrt{\sin^3 x - \sin^5 x} dx$

Since $\sqrt{\sin^3 x - \sin^5 x} = \sqrt{\sin^3 x (1 - \sin^2 x)} = \sin^{\frac{3}{2}} \cdot |\cos x|$, where $|\cos x| = \cos x$ on $[0, \frac{\pi}{2}]$; $|\cos x| = -\cos x$ on $[\frac{\pi}{2}, \pi]$,

$$\begin{aligned} & \int_0^\pi \sqrt{\sin^3 x - \sin^5 x} dx \\ &= \int_0^{\frac{\pi}{2}} \sin^{\frac{3}{2}} x \cos x dx + \int_{\frac{\pi}{2}}^\pi \sin^{\frac{3}{2}} x (-\cos x) dx \\ &= \int_0^{\frac{\pi}{2}} \sin^{\frac{3}{2}} x d(\sin x) - \int_{\frac{\pi}{2}}^\pi \sin^{\frac{3}{2}} x d(\sin x) \\ &= \left[\frac{2}{5} \sin^{\frac{5}{2}} x \right]_0^{\frac{\pi}{2}} - \left[\frac{2}{5} \sin^{\frac{5}{2}} x \right]_{\frac{\pi}{2}}^\pi \\ &= \frac{2}{5} - \left(-\frac{2}{5} \right) = \frac{4}{5} \end{aligned}$$

Solutions: 2. $\int_0^3 \frac{x^2}{(x^2 - 3x + 3)^2} dx$

$$x^2 - 3x + 3 = (x - \frac{3}{2})^2 + \frac{3}{4}, \text{ let } x - \frac{3}{2} = \frac{\sqrt{3}}{2} \tan u (|u| < \frac{\pi}{2}), \text{ such that}$$

$$(x^2 - 3x + 3)^2 = (\frac{3}{4} \sec^2 u)^2, dx = \frac{\sqrt{3}}{2} \sec^2 u du.$$

$$x = 0 \rightarrow u = -\frac{\pi}{3}; x = 3 \rightarrow u = \frac{\pi}{3}.$$

$$\int_0^3 \frac{x^2}{(x^2 - 3x + 3)^2} dx = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} (\frac{3}{4} \tan^2 u + \frac{3\sqrt{3}}{2} \tan u + \frac{9}{4}) \cdot \frac{16}{9} \cdot \frac{\sqrt{3}}{2} \cos^2 u du$$

$$= \frac{8}{3\sqrt{3}} \cdot 2 \int_0^{\frac{\pi}{3}} (\frac{3}{4} \tan^2 u + \frac{9}{4}) \cos^2 u du$$

$$= \frac{4}{\sqrt{3}} \int_0^{\frac{\pi}{3}} (\sin^2 u + 3 \cos^2 u) du = \frac{4}{\sqrt{3}} \int_0^{\frac{\pi}{3}} (2 + \cos 2u) du$$

$$= \frac{4}{\sqrt{3}} [2u + \frac{1}{2} \sin 2u]_0^{\frac{\pi}{3}}$$

$$= \frac{8\pi}{3\sqrt{3}} + 1$$

Definition

Type1: Unboundedness

- ① If $f(x)$ is continuous on $[a, +\infty]$, for any $t > a$, the improper integral of $f(x)$ on $[a, +\infty]$ is

$$\int_a^{+\infty} f(x) dx = \lim_{t \rightarrow +\infty} \int_a^t f(x) dx$$

- ② If $f(x)$ is continuous on $[-\infty, a]$, for any $t < a$, the improper integral of $f(x)$ on $[-\infty, a]$ is

$$\int_a^{+\infty} f(x) dx = \lim_{t \rightarrow -\infty} \int_t^a f(x) dx$$

Definition

Type1: Unboundedness

- ③ If both $\int_a^{\infty} f(x)dx$ and $\int_{-\infty}^a f(x)dx$ are convergent

$$\int_{-\infty}^{+\infty} f(x)dx = \int_a^{+\infty} f(x)dx + \int_{-\infty}^a f(x)dx$$

Definition

Type2: Discontinuous

- ① If f is continuous on $[a, b)$ and is discontinuous at b , then the improper integral on is

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

- ② If f is continuous on $(a, b]$ and is discontinuous at a , then the improper integral on is

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$$

Definition

Type2: Discontinuous

- ③ If f has a discontinuity at c ($a < c < b$), and both $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ are convergent, then

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

Convergence and divergence of improper integrals

If the limits mentioned above exist (as a finite number), then the improper integral is called convergent, otherwise it's divergent.

$$\textcircled{1} \int_e^\infty \frac{1}{x(\ln x)^3} dx$$

$$\textcircled{2} \int_0^3 \frac{1}{x^2 - 6x + 5} dx$$

Solutions: 1.

$$\begin{aligned}\int_e^\infty \frac{1}{x(\ln x)^3} dx &= \lim_{t \rightarrow \infty} \int_e^t \frac{1}{x(\ln x)^3} dx \\ &= \lim_{t \rightarrow \infty} \int_1^{\ln t} u^{-3} du\end{aligned}$$

($u = \ln x$, $du = dx/x$)

$$\begin{aligned}&= \lim_{t \rightarrow \infty} \left[-\frac{1}{2u^2} \right]_1^{\ln t} \\ &= \lim_{t \rightarrow \infty} \left[-\frac{1}{2(\ln t)^2} + \frac{1}{2} \right] \\ &= 0 + \frac{1}{2} = \frac{1}{2}\end{aligned}$$

Convergent

Solutions:

$$2. \int_0^3 \frac{dx}{x^2 - 6x + 5} = \int_0^3 \frac{dx}{(x-1)(x-5)}$$

$$= \int_0^1 \frac{dx}{(x-1)(x-5)} + \int_1^3 \frac{dx}{(x-1)(x-5)}$$

$$= I_1 + I_2$$

$$I_1 = \frac{1}{4} \lim_{t \rightarrow 1^-} \int_0^t \left(\frac{1}{x-5} - \frac{1}{x-1} \right) dx$$

$$= \lim_{t \rightarrow 1^-} \left[-\frac{1}{4} \ln|t-1| + \frac{1}{4} \ln|t-5| \right]_0^t$$

$$= \lim_{t \rightarrow 1^-} \left[\left(-\frac{1}{4} \ln|t-1| + \frac{1}{4} \ln|t-5| \right) - \left(-\frac{1}{4} \ln|-1| + \frac{1}{4} \ln|-5| \right) \right]_0^t$$

$$= \infty$$

Since I_1 is divergent, I is divergent.

Theorem 1

Suppose $f(x)$ is continuous on $[a, \infty)$, and $f(x) \geq 0$. If

$$F(x) = \int_a^x f(t)dt$$

has an upper bound on $[a, \infty)$, then the improper integral $\int_a^\infty f(x)dx$ converges.

Theorem 2: Comparison Theorem

Suppose that f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.

- (a) If $\int_a^\infty f(x)dx$ is convergent, then $\int_a^\infty g(x)dx$ is convergent.
- (b) If $\int_a^\infty g(x)dx$ is divergent, then $\int_a^\infty f(x)dx$ is divergent.

Judge whether the following improper integrals converge or not. If some of them converge, calculate their value.

$$\textcircled{1} \int_1^{\infty} \frac{1}{x^4} dx$$

$$\textcircled{2} \int_0^2 \frac{dx}{(1-x)^2}$$

$$\textcircled{3} \int_1^{\infty} \frac{1}{\sqrt{x}} dx$$

$$\textcircled{4} \int_0^1 \frac{x}{\sqrt{1-x^2}} dx$$

Solutions:

$$\textcircled{1} \int_0^{\infty} \frac{dx}{x^4} = - \left[\frac{1}{3x^3} \right] = \frac{1}{3}$$

$$\textcircled{2} \int_0^t \frac{dx}{(1-x)^2} = \left[\frac{1}{1-x} \right]_0^t = \frac{1}{1-t} - 1 \text{ Diverge}$$

$$\textcircled{3} \int_1^t \frac{1}{\sqrt{x}} dx = [2\sqrt{x}]_1^t = 2\sqrt{t} - 2 \text{ Diverge}$$

$$\textcircled{4} \int_0^1 \frac{xdx}{1-x^2} = -[\sqrt{1-x^2}]_0^1 = 1$$

Calculate the following improper integral:

$$\int_0^{\infty} \frac{1 - e^{-x^2}}{x^2} dx$$

You can directly use the conclusion that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.
(How to calculate this integral will be taught in VV255.)

Solutions:

$$\int_0^{\infty} \frac{1 - 3^{-x^2}}{x^2} dx = \int_0^{\infty} (e^{-x^2} - 1) d\left(\frac{1}{x}\right)$$

$$= \left[\frac{e^{-x^2} - 1}{x} \right]_{\infty}^0 - \int_0^{\infty} \frac{1}{x} d(e^{-x^2} - 1)$$

$$= \left[\frac{e^{-x^2} - 1}{x} \right]_0^{\infty} + 2 \int_0^{\infty} e^{-x^2} dx$$

$$\lim_{x \rightarrow \infty} \frac{e^{-x^2} - 1}{x} = 0, \lim_{x \rightarrow 0} \frac{e^{-x^2} - 1}{x} = \lim_{x \rightarrow 0} \frac{-2xe^{-x^2}}{1} = 0$$

$$2 \int_0^{\infty} e^{-x^2} dx = \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$\text{Therefore, } \int_0^{\infty} \frac{1 - 3^{-x^2}}{x^2} dx = \sqrt{\pi}$$

Theorem

$$\frac{d}{dt} \int_{\alpha(t)}^{\beta(t)} f(x) dx = f(\beta(t))\beta'(t) - f(\alpha(t))\alpha'(t)$$

How to prove it?

Use the integration to deal with the combination of limits and series.

Example

$$① \lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{i}{n^2}$$

$$② \lim_{n \rightarrow \infty} \sqrt{n} \left(\frac{1}{n} - \sum_{i=0}^n \frac{1}{n + \sqrt{i}} \right)$$

Method

① Simply Transform into Integration. $\lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{i}{n^2} = \int_0^n \frac{i}{n^2} di = \frac{1}{2}$

② Use Squeeze Theorem. $ans = \lim_{n \rightarrow \infty} \sqrt{n} \sum_{i=0}^n \frac{\sqrt{i}}{n(n + \sqrt{i})}$

$$ans \leq \lim_{n \rightarrow \infty} \sqrt{n} \sum_{i=0}^n \frac{\sqrt{i}}{n^2} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^n \sqrt{\frac{k}{n}} = \int_0^1 \sqrt{x} dx = \frac{2}{3}.$$

$$ans \geq \lim_{n \rightarrow \infty} \sqrt{n} \sum_{i=0}^n \frac{\sqrt{i}}{n(n + \sqrt{n})} = \lim_{n \rightarrow \infty} \frac{n}{n + \sqrt{n}} \frac{1}{n} \sum_{i=0}^n \sqrt{\frac{k}{n}} = \frac{2}{3}.$$

Exercise: Refer to Worksheet 1 1.3,1.4

- [1] Huang, Yucheng. VV156_RC2.pdf. 2021.
- [2] Cai, Runze. Chapter02.pdf. 2021.
- [3] Department of mathematics, Tongji University. Advanced Mathematics (7th Edition). 2014.
- [4] James Stewart. Calculus (7th Edition). 2014.
- [5] Department of mathematics, Tongji University. Learning Guidance of Advanced Mathematics (7th Edition). 2014.
- [6] Cai, Runze. Chapter03.pdf. 2021.
- [7] Huang, Yucheng. VV156_RC3.pdf. 2021.
- [8] Chen, Jixiu et al. Mathematical Analysis (3rd Version). 2019
- [9] Li, Junhao. VV156 Regular RC3/RC4.pdf. 2022.
- [10] Zhou, Yishen. VV156 Mid2 Big RC Part1.pdf. 2022.

Any Questions?

